

IE5203 Decision Modeling & Risk Analysis

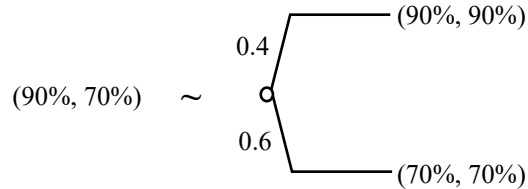
Solutions to Chapter 10 Exercises

P10.1

(a) Given $u(x_1, x_2) = k_1 u_1(x_1) + (1 - k_1) u_2(x_2)$

$$70\% \leq x_1 \leq 90\% \text{ and } 70\% \leq x_2 \leq 90\%.$$

$$\begin{aligned} \text{Let } u(70\%, 70\%) &= u_1(70\%) = u_2(70\%) = 0 \\ u(90\%, 90\%) &= u_1(90\%) = u_2(90\%) = 1 \end{aligned}$$

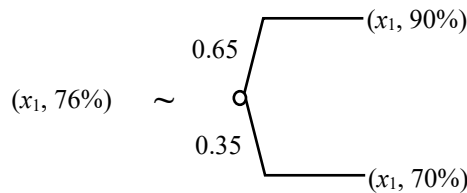


$$\begin{aligned} \Rightarrow 0.4 u(90\%, 90\%) + 0.6 u(70\%, 70\%) &= u(90\%, 70\%) \\ 0.4 u(90\%, 90\%) + 0.6 u(70\%, 70\%) &= k_1 u_1(90\%) + (1 - k_1) u_2(70\%) \\ 0.4 (1) + 0.6 (0) &= k_1 (1) + (1 - k_1) (0) \end{aligned}$$

$$\Rightarrow k_1 = 0.4$$

- Hence the Ministry two-attribute additive utility function is $u(x_1, x_2) = 0.4 u_1(x_1) + 0.6 u_2(x_2)$

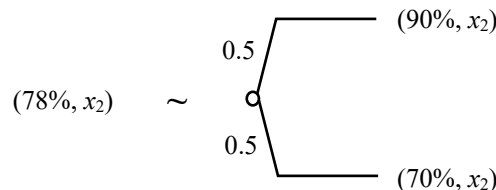
(b)



$$\Rightarrow 0.65 [k_1 u_1(x_1) + k_2 u_2(90\%)] + 0.35 [k_1 u_1(x_1) + k_2 u_2(70\%)] = k_1 u_1(x_1) + k_2 u_2(76\%)$$

$$\underline{k}_1 u_1(x_1) + 0.65 k_2 u_2(90\%) + 0.35 k_2 u_2(70\%) = \underline{k}_1 u_1(x_1) + k_2 u_2(76\%)$$

$$\text{Hence } \underline{u}_2(76\%) = 0.65.$$

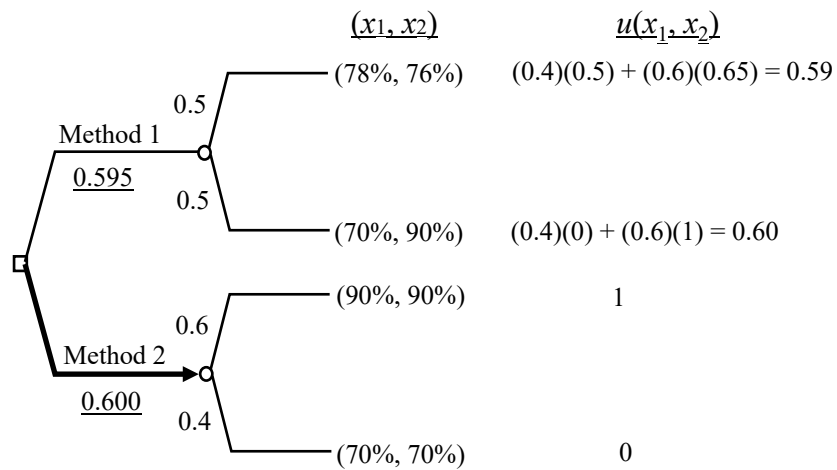


$$\Rightarrow 0.5 [k_1 u_1(90\%) + k_2 u_2(x_2)] + 0.5 [k_1 u_1(70\%) + k_2 u_2(x_2)] = k_1 u_1(78\%) + k_2 u_2(x_2)$$

$$0.5 k_1 u_1(90\%) + k_2 u_2(x_2) + 0.5 k_1 u_1(70\%) = k_1 u_1(78\%) + k_2 u_2(x_2)$$

$$\text{Hence } u_1(78\%) = 0.5.$$

- The decision tree for the two teaching techniques is as follows:



Conclusion: Method 2 is preferred to Method 1.

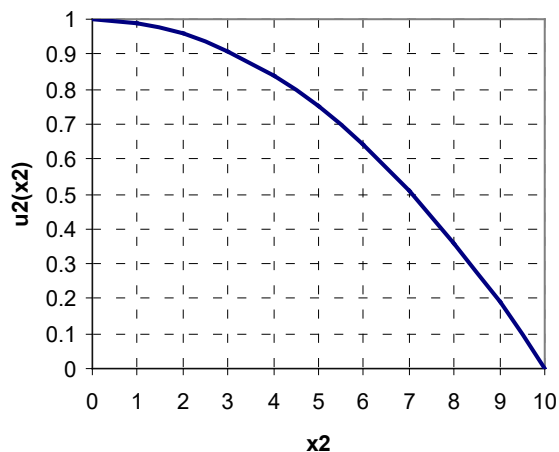
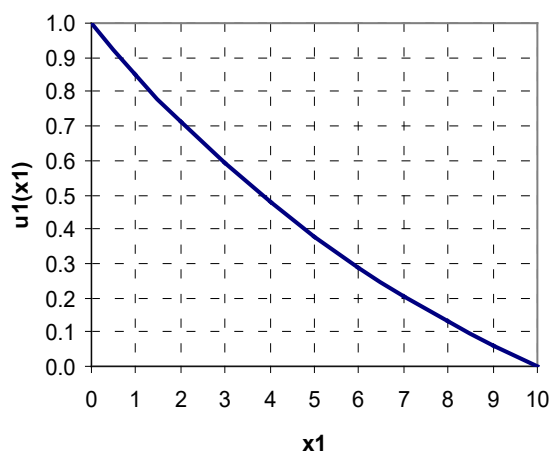
P10.2

x_1 = Weekly blood shortage ($0 \leq x_1 \leq 10$)

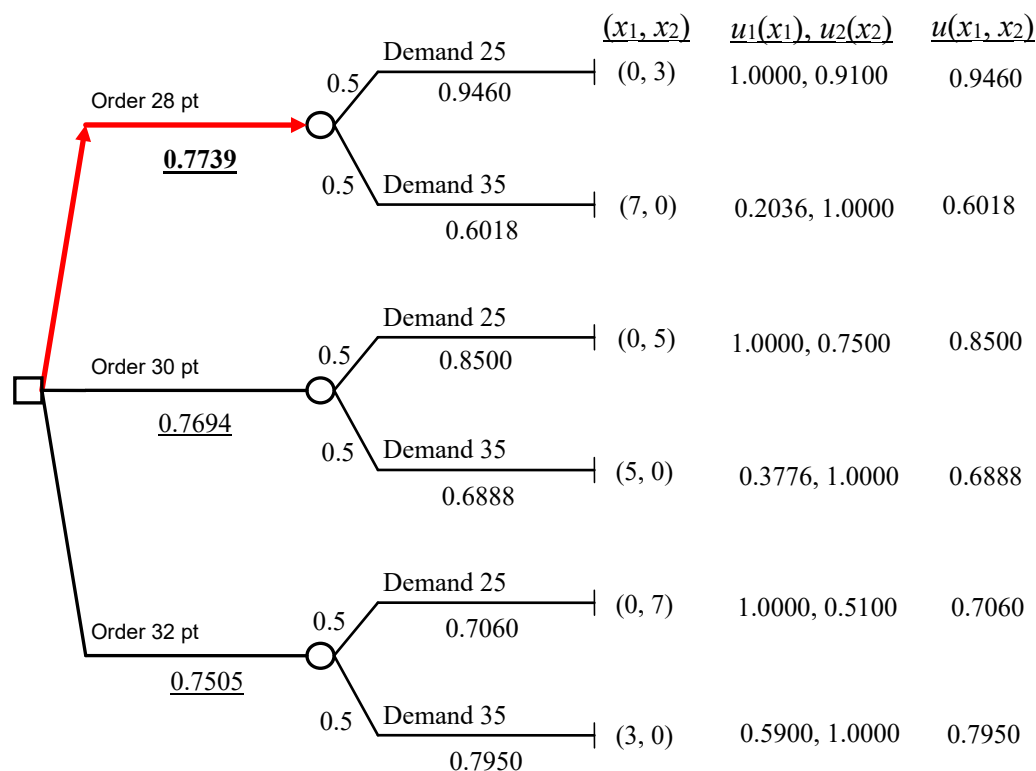
x_2 = Weekly blood outdated ($0 \leq x_2 \leq 10$)

$$u(x_1, x_2) = 0.4 u_1(x_1) + 0.5 u_2(x_2) + 0.1 u_1(x_1) u_2(x_2)$$

where $u_1(x_1) = 0.582 \left[\exp\left(1 - \frac{x_1}{10}\right) - 1 \right]$ and $u_2(x_2) = 1 - \frac{x_2^2}{100}$.



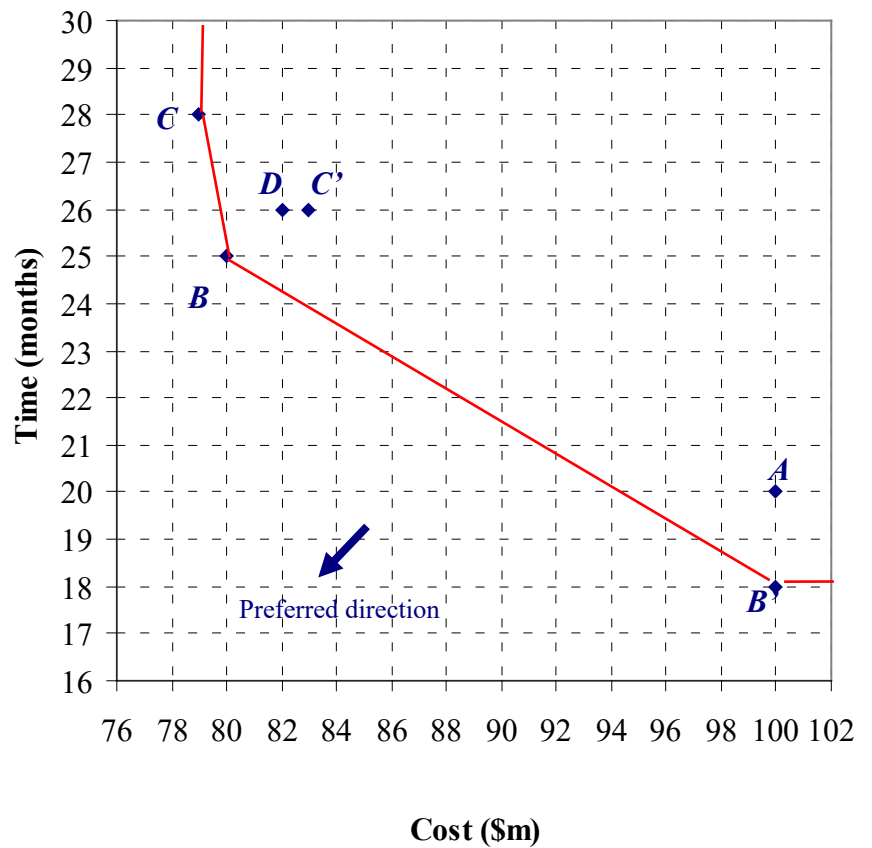
The decision tree is



Conclusion: The hospital should order 28 pints of blood weekly.

P10.3

Contractor	Cost (\$m)	Time (months)
<i>A</i>	100	20
<i>B</i>	80	25
<i>B'</i>	100	18
<i>C</i>	79	28
<i>C'</i>	83	26
<i>D</i>	82	26



Given $B' \sim B$ and $C' \sim C$.

By dominance analysis, we have:

$B \succ D$, $D \succ C'$ and $B' \succ A$

Hence $B' \sim B \succ D \succ C' \sim C$
and $B' \succ A$.

Answer: Choose either B or B' .

Note that although C is an efficient alternative, i.e., on the efficient frontier, it is not optimal to the decision maker because it has the same utility as C' , which is non-efficient.

Hence, non-dominance is necessary but not sufficient for optimality with respect to utility.

P10.4

(a) The weights for the main criteria with respect to the Goal are computed:

	Human prod	Economics	Design	Operations	Exact w	RGM
Human Productivity	1	3	3	7	0.513052	0.5159
Economics	1/3	1	2	5	0.246592	0.2474
Design	1/3	1/2	1	7	0.193575	0.1903
Operations	1/7	1/5	1/7	1	0.046781	0.0463

$$\lambda_{\max} = 4.212088, \text{ CR} = 0.078551 < 10\%$$

The local weights for the alternatives with respect to each criterion are computed:

Human Productivity:

	System A	System B	System C	Exact w	RGM
System A	1	3	5	0.648329	0.6483
System B	1/3	1	2	0.229651	0.2297
System C	1/5	1/2	1	0.122020	0.1220

$$\lambda_{\max} = 3.003695, \text{ CR} = 0.003185 < 10\%$$

Economics:

	System A	System B	System C		RGM
System A	1	1/3	1/2	0.157056	0.1571
System B	3	1	3	0.593634	0.5936
System C	2	1/3	1	0.249311	0.2493

$$\lambda_{\max} = 3.053622, \text{ CR} = 0.046225 < 10\%$$

Design:

	System A	System B	System C	Exact w	RGM
System A	1	1/2	1/7	0.093813	0.0938
System B	2	1	1/5	0.166593	0.1666
System C	7	5	1	0.739594	0.7396

$$\lambda_{\max} = 3.014152, \text{ CR} = 0.012200 < 10\%$$

Operations:

	System A	System B	System C	Exact w	RGM
System A	1	3	1/5	0.178178	0.1782
System B	1/3	1	1/9	0.070418	0.0704
System C	5	9	1	0.751405	0.7514

$$\lambda_{\max} = 3.029064, \text{ CR} = 0.025055 < 10\%$$

The composite weights for the three alternative systems are:

- System A: $(0.513052)(0.648329) + (0.246592)(0.157056) + (0.093813)(0.0938) + (0.046781)(0.178178)$
 $= 0.397850$
- System B: $(0.513052)(0.229651) + (0.246592)(0.593634) + (0.193575)(0.166593) + (0.046781)(0.070418)$
 $= 0.299750$
- System C: $(0.513052)(0.122020) + (0.246592)(0.249311) + (0.193575)(0.739594) + (0.046781)(0.751405)$
 $= 0.302399$

Hence System A should be chosen as it has the highest global weight.

(b)

Alternative	Effectiveness	EUAC (\$)
System B	0.299750	80,000
System A	0.397850	100,000
System C	0.302399	110,000

- Efficient Cost-Effective Alternatives are B and A.
- The efficient frontier is shown below:

